

A Geometry-Only Tension Probe for Reverse Exploration of Single-Block SHA-256

Exact one-step reverse algebra, a zero-score theorem for true schedule words, and empirical evidence of a nontrivial hot/cold field

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Proof boundary

This paper proves exact one-step reverse closure conditioned on $W[t]$, proves a zero-score theorem for the true word under the exported geometry bundle, and reports empirical hot/cold behavior. It does not claim a proof of SHA-256 inversion.

Abstract

This paper formalizes and checks the mathematics behind a one-round reverse probe for single-block SHA-256. The probe is explicitly separated into two logically distinct parts: (i) a geometry-only hot/cold score that uses admissible side data only, and (ii) a hidden verifier that knows the true schedule word and exists only for evaluation. The reverse algebra is exact once the next state $x[t+1]$ and the candidate schedule word $W[t]$ are given. We prove an exact one-step reverse closure theorem under that condition, prove a zero-score theorem for the true word under the exported side-observable bundle, and show why these statements do not amount to a proof of SHA-256 inversion. Empirically, the companion notebook shows a strong gradient: for the demo message `b"abc"`, the true $W[63] = 0x12b1edeb$ receives score 0 while sampled false guesses score substantially higher on average; across six test messages the true word ranked first among 5,000 sampled candidates in each trial. At the same time, low-scoring false positives remain, so the present construction is best interpreted as a tension field rather than a full reverse oracle.

1. Introduction

The motivating intuition is mechanical rather than mystical. A lock picker does not know the correct pin setting in advance; instead, they apply tension and read which candidate settings bind cleanly and which ones fight the mechanism. The present SHA-256 probe is built to test whether an analogous effect can be extracted from admissible side geometry during a reverse walk. The question is not yet “can we invert SHA-256?” but a narrower one: “if we guess a schedule word in the reverse direction, does the internal addition geometry supply a measurable hot/cold signal?”

To keep the logic clean, the construction forbids direct transport values in the visible game state. The exported bundle contains only the digest and a small family of per-round observables derived from the forward pass: staged carry-out bits, staged carry-mask Hamming weights, and the Hamming weight of the h-register entering the T1 chain. No $W[t]$, no padded message words, and no round-state words are exposed in the geometry bundle. A second, separate verifier stores only salted commitments to the true words so that the geometry probe can be evaluated without being conflated with an answer-key oracle.

The companion script and executed notebook that accompany this paper implement the construction and provide the empirical data summarized below. The paper’s aim is to state exactly what the mathematics proves, what the experiments demonstrate, and what remains open.

2. Formal Model

Fix a single 512-bit SHA-256 message block. Let

$$x_t = [a_t, b_t, c_t, d_t, e_t, f_t, g_t, h_t]^T \in ((\mathbb{Z}/2^{32}\mathbb{Z}))^8,$$

and let the round recurrence be

$$x_{t+1} = P x_t + u_a(T1_t + T2_t) + u_e T1_t \pmod{2^{32}}, \quad t = 0, \dots, 63,$$

where P is the shift matrix and

$$T1_t = h_t + \Sigma1(e_t) + Ch(e_t, f_t, g_t) + K_t + W_t \pmod{2^{32}},$$

$$T2_t = \Sigma0(a_t) + Maj(a_t, b_t, c_t) \pmod{2^{32}}.$$

Only two active injections occur per round:

$$a_{t+1} = T1_t + T2_t \pmod{2^{32}}, \quad e_{t+1} = d_t + T1_t \pmod{2^{32}},$$

while the remaining lanes are pure shifts.

For bit-level side geometry we also use two exact primitives from the implementation:

1. $\text{add32}(a, b)$ returns $(a+b \pmod{2^{32}}, \text{carry_out}, \text{carry_mask})$, where carry_out is the final overflow bit and carry_mask is the closure of all internal carry positions generated during the two-word addition.
2. $\text{hw}(z)$ denotes the Hamming weight of a 32-bit word z .

3. Exact One-Step Reverse Algebra

Given $x[t+1]$ and a candidate value for $W[t]$, most of $x[t]$ is fixed immediately by the shift structure:

$$a_t = b_{t+1}, b_t = c_{t+1}, c_t = d_{t+1}, e_t = f_{t+1}, f_t = g_{t+1}, g_t = h_{t+1} .$$

Then $T_2[t]$ is determined exactly from $a[t], b[t], c[t]$:

$$T_2_t = \Sigma_0(a_t) + \text{Maj}(a_t, b_t, c_t) \pmod{2^{32}} .$$

Since $a[t+1] = T_1[t] + T_2[t]$, one recovers

$$T_1_t = a_{t+1} - T_2_t \pmod{2^{32}} .$$

Since $e[t+1] = d[t] + T_1[t]$, one then recovers

$$d_t = e_{t+1} - T_1_t \pmod{2^{32}} .$$

Finally, because

$$T_1_t = h_t + \Sigma_1(e_t) + \text{Ch}(e_t, f_t, g_t) + K_t + W_t \pmod{2^{32}} ,$$

the candidate $W[t]$ determines $h[t]$ exactly via

$$h_t = T_1_t - \Sigma_1(e_t) - \text{Ch}(e_t, f_t, g_t) - K_t - W_t \pmod{2^{32}} .$$

This yields a complete candidate predecessor state $x[t]$ for each guessed $W[t]$.

4. Geometry Bundle and Tension Score

For each round t the visible geometry bundle exports only the tuple

$$G_t = (c_t^{(1)}, c_t^{(2)}, c_t^{(3)}, c_t^{(4)}; h_1, h_2, h_3, h_4; w_t),$$

where the four c -values are the staged carry-out bits produced by the chained additions

$$h_t + \Sigma_1(e_t), (+\text{Ch}), (+K_t), (+W_t),$$

the four h -values are the Hamming weights of the corresponding carry masks from those same additions, and $\omega[t] = \text{hw}(h[t])$ is the Hamming weight of the reconstructed h -register. This bundle intentionally excludes the direct transport value $W[t]$ and the full round state.

Given a candidate guess g for $W[t]$, the reverse map produces a predicted geometry tuple $G_t(g)$. The geometry-only tension score is then

$$S_t(g) = 5 \sum_{i=1}^4 1[c_t^{(i)}(g) \neq c_t^{(i)}] + \sum_{i=1}^4 |h_i(g) - h_i| + |w_t(g) - \omega_t| .$$

The weighting factor 5 on carry-bit mismatches is an implementation choice designed to make carry disagreements more expensive than small Hamming-weight drift. The score is lower-is-better. In the companion notebook the qualitative labels are calibrated from random sampling: score 0 is labeled "ice," then progressively larger quantiles map to cold, cool, warm, hot, and burning.

5. Statements and Proofs

Theorem 1 (Exact one-step reverse closure conditioned on $W[t]$).

Fix a round t and a next state $x[t+1]$. For any candidate schedule word g in $\mathbb{Z}/2^{32}\mathbb{Z}$, the formulas above reconstruct a unique candidate predecessor state $x[t](g)$. If g equals the true $W[t]$ generated in the forward run, then $x[t](g)$ equals the true predecessor state.

Proof.

The shift equations determine six coordinates of $x[t]$ exactly. $T_2[t]$ is then a deterministic function of $a[t]$, $b[t]$, $c[t]$. The equation $a[t+1] = T_1[t] + T_2[t]$ gives $T_1[t]$ uniquely modulo 2^{32} , and $e[t+1] = d[t] + T_1[t]$ gives $d[t]$ uniquely modulo 2^{32} . Finally, the definition of $T_1[t]$ is affine in $h[t]$ once $e[t]$, $f[t]$, $g[t]$, $K[t]$, and the candidate g are fixed, so $h[t]$ is uniquely determined modulo 2^{32} . Substituting the true $W[t]$ reproduces the original defining equations of the forward round, hence recovers the true predecessor state. $\square$

Theorem 2 (Zero-score theorem for the true word).

Let $G[t]$ be the side-observable tuple exported from the forward pass at round t . If $g = W[t]$ is the true schedule word for that round, then $S[t](g) = 0$.

Proof.

By Theorem 1, substituting the true $W[t]$ into the reverse formulas reconstructs the exact predecessor state $x[t]$. Re-running the staged additions $h[t] + \Sigma_1(e[t])$, then $+Ch$, then $+K[t]$, then $+W[t]$, necessarily reproduces the same carry-out bits, the same carry masks, and the same h -register word as in the forward pass, because the operands are identical at each stage. Therefore every carry-bit mismatch term is zero, every carry-mask Hamming-weight difference is zero, and the h -register Hamming-weight difference is zero. Hence $S[t](W[t]) = 0$. $\square$

Proposition 3 (Logical separation of geometry probe and verifier).

The geometry score $S[t](g)$ can be computed from the visible bundle and a candidate guess without directly comparing g to the true $W[t]$. The hidden verifier, by contrast, is an answer-key oracle implemented via salted commitments to the true words.

Proof.

Inspection of the implementation shows that `geometry_score()` uses only the visible round bundle, the next-state digest-derived working state, and the candidate guess. The function never reads the stored true word or its commitment. The verifier probe checks equality only through commitment comparison. The two procedures are therefore logically distinct. $\square$

Corollary 4 (What is and is not proved).

The construction proves an exact one-step reverse map conditioned on a candidate schedule word and proves that the true word obtains score zero. It does not prove that score zero or low score uniquely identifies the true word over the full 32-bit space.

Reason.

Uniqueness would require a theorem that the exported side-observable tuple is injective in $W[t]$ for fixed $x[t+1]$. No such theorem is established here, and the experiments below exhibit low-scoring false positives. $\square$

6. Experimental Protocol

The experiments use the companion implementation `sha_reverse_tension_probe.py` and its executed notebook. All tests are restricted to single-block messages of length at most 55 bytes, so that the SHA-256 padding remains within one 512-bit block.

For each message, the procedure is:

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github.com/QuHarmonics/The-Nexus-Harmonic-Reality

1. Run the full forward pass and collect the per-round geometry bundle.
2. Recover the working final state from the digest by subtracting the initial hash vector H_0 componentwise modulo 2^{32} .
3. At round $t = 63$, evaluate the true word and several handpicked false guesses.
4. Calibrate the hot/cold labels from 1,024 random 32-bit samples.
5. Sample 5,000 random guesses and rank them by the geometry-only score.
6. Compare against the hidden verifier to determine whether the true word is unusually favored by the geometry probe.

7. Results

7.1 Demo Result for $m = b"abc"$

Digest: ba7816bf8f01cfea414140de5dae2223b00361a396177a9cb410ff61f20015ad

First reverse target: $t = 63$

True hidden word: $W[63] = \text{ox12b1edeb}$

Calibration quantiles: min=5, q05=12, q20=19, median=26, q80=34, max=56

Guess	Score	Label	Carry mismatch	Mask-HW error	h-HW error	Verified
ox12b1edeb	0	ice	0	0	0	True
ox00000000	26	warm	0	23	3	False
oxffffff	26	warm	1	19	2	False
ox12345678	31	hot	0	26	5	False

In the 5,000-sample random search, the true word ranked #1. The five best false candidates and their scores were:

False guess	Score
ox185251bd	4
ox1ebf3deb	4
ox1fb9ec2c	4
ox0eabf949	5
ox0ff0e2d3	5

In a separate 100,000-sample random scan for $m = b"abc"$, no false zero-score sample appeared. The minimum false score observed was 1 at guess ox1f0f63e2; 103 false samples scored ≤ 4 , 239 scored ≤ 5 , and 3031 scored ≤ 10 .

7.2 Multi-Message Sampling Results

Message	Len	Digest prefix	True $W[63]$	True score	True rank	Median cal	q05 cal	Best false score	Best false guess	False ≤ 5
b'abc'	3	ba7816bf8f01cfea	ox12b1edeb	0	1	26	12	4	ox185251bd	10

Message	Le n	Digest prefix	True W[63]	True score	True rank	Media n cal	qo 5 cal	Best false score	Best false guess	Fals e ≤5
b'hello'	5	2cf24dba5fbo30e	ox3e3cb561	0	1	29	13	2	ox52b9bf39	16
b'a'	1	ca978112ca1bbdc	ox1a1973f6a	0	1	22	12	3	ox16f35ab9	9
b''	0	e3b0c44298fc1c14	ox3b5ec49b	0	1	27	10	1	ox3baf3c5a	21
b'OpenAI'	6	8b7d1a3187ab355d	oxa0876f01	0	1	31	12	1	oxaa211b2b	19
b'\\x00..\\x0f'	16	be45cb2605bf36be	ox07e63711	0	1	25	11	4	ox3f8f3fo6	9

In all six trials the true word ranked first among 5,000 sampled candidates, but low-scoring false positives remained.

8. Discussion

The experimental picture is consistent across all tested messages. The true schedule word always receives score zero, as Theorem 2 requires. In random-sample rankings, the true word ranked first in all six 5,000-sample trials performed for this paper. This is strong evidence that the score function is not decorative: it is measuring a real gradient in the side-data manifold.

At the same time, the probe is not unique at one round. In the b"abc" experiment, three false guesses scored 4 and several more scored 5 in the 5,000-sample search. In a 100,000-sample random scan for the same message, no false zero-score sample appeared, but the minimum false score observed was 1. Therefore the present probe should be described as a tension meter, not a proof of recovery of W[63] from side geometry alone.

The lock-picking analogy is therefore mathematically apt but must be stated carefully. Tension is present: some guesses bind cleanly and some do not. Yet a single pin setting has not opened the lock. A local cold score indicates compatibility with the exported constraints, not global correctness.

9. Limitations

First, the construction is restricted to one round and one block. The paper does not prove that the same geometry bundle remains sufficiently discriminative when coupled across multiple reverse rounds, although that is the natural next experiment.

Second, the visible bundle is deliberately thin. It exports only four carry bits, four carry-mask Hamming weights, and one register Hamming weight per round. A richer but still admissible bundle might sharpen discrimination. Conversely, one must be careful not to leak transport values by disguising them as "geometry."

Third, the random-sampling results are empirical only. Ranking first among 5,000 random candidates does not imply anything like a polynomial-time inversion theorem for SHA-256. The present work establishes a measurable local gradient and nothing stronger.

Fourth, the proof boundary is exact but conditional. The mathematics proves exact reverse closure once $W[t]$ is specified. It does not remove the central challenge of finding the correct $W[t]$ without an oracle.

10. Conclusion

This paper checks and formalizes a narrow but important claim: a geometry-only one-round reverse probe for single-block SHA-256 can be built, its reverse algebra is exact, the true schedule word provably scores zero, and the resulting score function empirically behaves like a hot/cold tension field rather than a flat random labeler.

The work does not prove SHA-256 inversion. It does, however, replace the strictly blind local guess with a constrained local guess that can be ranked by admissible side geometry. In that sense the problem has shifted from “guess and hope” to “guess under tension.”

The immediate next step is multi-round coupling. If a candidate word must remain cold not for one round but for a chained sequence of reverse steps, the false-positive basin should tighten. Whether that tightening becomes strong enough to support practical recovery remains an open research question.

Appendix A. Reproducibility Summary

Algorithm 1 (One-round geometry probe):

Input: message m of length ≤ 55 bytes, round index $t = 63$, candidate guess g .

Output: geometry score $S[t](g)$, optional verifier answer.

A1. Pad m to a single SHA-256 block and expand the schedule $W[0..63]$.

A2. Run the forward compression loop and collect, for each round, the four staged carry bits, the four carry-mask Hamming weights, and $hw(h[t])$.

A3. Compute the digest and derive the working final state by subtracting H_0 modulo 2^{32} .

A4. Given guess g , reconstruct $x[t]$ from $x[t+1]$ using the reverse formulas above.

A5. Re-run the staged T_1 chain on the reconstructed state and measure the three mismatch components.

A6. Return $S[t](g) = 5 \cdot \text{carries} + \text{mask drift} + \text{h drift}$.

A7. If evaluation is desired, compare g to the hidden commitment-based verifier.

The companion notebook, executed notebook, and script are the definitive computational appendix for this paper.

References

[1] Companion script: `sha_reverse_tension_probe.py`.

[2] Companion notebook: `sha_reverse_tension_probe.ipynb`.

[3] Executed notebook: `sha_reverse_tension_probe.executed.ipynb`.